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Title / Título

PROGRAM ANCOLITE USER MANUAL

Keywords / Palabras clave	Access class Clasificación acceso
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Summary / Resumen

This document is the user's manual of program *ANCOlite*, version 1.0.

This program calculates the stiffness matrices and apparent/equivalent Young moduli of fibrous composite laminates according to the *Classical Laminate Theory*, as well as:

- the strain levels seen in the laminate under in-plane loads,
- the uni-axial buckling loads of a simply supported rectangular plate/shell made of that laminate, and,
- generates the *Nastran BDF* entries (*MAT8, PCOMP* and *MAT2*) required to use that laminated material in Finite Elements Analyses with that solver

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Structural Analysis	**Author** **Autor**	Gustavo Baños Lapuente	**Signature** **Firma**
Stress Methods and Optimisation			Electronic signature
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	Authorisation **Autorización**		

REVISIONS RECORD / REGISTRO DE REVISIONES

Issue / Revisión	**Change reason / Motivo de Modificación**	**Author / Autor**	**Tech. Approval / Aprob. Técnica**	**Authorisation / Autorización**
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1 INTRODUCTION

1.1 TECHNICAL SHEET OF THE PROGRAM

Technical Report: TEA-T-NT-210037

PROGRAM	DATE
ANCOlite	06/2021

OBJECTIVE

ANCOlite calculates the stiffness matrices and apparent/equivalent Young moduli of fibrous composite laminates according to the *Classical Laminate Theory*

LANGUAGE	DEVELOPMENT TOOLS	OS/PLATFORM
Fortran 2003	MS VS 2015 + Intel Fortran 2018	Windows (x64)
Visual Basic Net	MS Visual Studio 2015	Windows (x64)

NUMBER OF LINES (APPROX.)	EXECUTION MODE
9000 (Fortran, analysis core) (*)	
+ 1000 (Visual Basic .Net)	Interactive GUI
Total 10000	

Notes:

- Blank lines and commented lines excluded
- Count for Visual Basic code excludes lines created by Visual Studio in designer.vb files
- A significant part of the Fortran analysis core is directly copied from *SAMFOR* project (status at June 2021) (*), and many code lines and functionalities there are not actually used by *ANCOlite*

AUTHOR	VERSION
Gustavo Baños	1.0

ADDITIONAL DOCUMENTS

See list of applicable references in chapter §1.4.

1.2 SOFTWARE PURPOSE AND DESCRIPTION

This document is the user's manual of program *ANCOlite*, version 1.0.

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- the strain levels seen in the laminate under in-plane loads,
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1.3 ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
BDF		Bulk Data File (input format for the *MSC Nastran* FEA solver
CLT		Classical Laminate Theory
FEA		Finite Elements Analysis
GUI		Graphical User Interface
MSVS		Microsoft Visual Studio 2015
OS		Operating System
CPT	mm	Cured Ply Thickness (thickness of the ply)
E_1	MPa	Young modulus of the ply in longitudinal direction
E_2	MPa	Young modulus of the ply in transverse direction
G_{12}	MPa	In-plane shear modulus of the ply
v_{12}	-	Poisson's ratio of the ply
α_1	1/ºC	Thermal expansion coefficient of the ply in longitudinal direction
α_2	1/ºC	Thermal expansion coefficient of the ply in transverse direction
t	mm	Thickness of the laminate (or ply, from context)
E_x	MPa	Young modulus of the laminate in X direction (0º in material reference axis)
E_y	MPa	Young modulus of the laminate in Y direction
G_{xy}	MPa	In-plane shear modulus of the laminate
N_{xy}	-	Poisson's ratio of the laminate
α_x	1/ºC	Thermal expansion coefficient of the laminate in X direction
α_y	1/ºC	Thermal expansion coefficient of the laminate in Y direction
α_{xy}	1/ºC	Thermal coupling coefficient of the laminate in the plane
[A]	N/mm	In-plane stiffness matrix of the laminate
[B]	N	Bending-membrane coupling matrix of the laminate
[D]	Nmm	Bending stiffness matrix of the laminate

N	N/mm	In-plane force per unit width
ε	-	Strain
bu	-	As subscript, refers to buckling onset
x	-	As subscript, refers to X direction on the laminate local reference axis
y	-	As subscript, refers to Y direction on the laminate local reference axis
xy	-	As subscript, refers to in-plane shear
L_x	mm	Length of the rectangular plate/shell
L_y	mm	Width of the rectangular plate/shell
R	mm	Radius of the shell (curvature on Y direction)

1.4 REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[Airbus DS] SMM	06/2021	Stress Methods Manual
Ref. 2	[Robert M. Jones]	2nd Ed.	MECHANICS OF COMPOSITE MATERIALS
Ref. 3	[SAE] CMH-17	07/12	COMPOSITE MATERIALS HANDBOOK
Ref. 4	ESDU 94007	-	ELASTIC BUCKLING OF CYLINDRICALLY CURVED LAMINATED FIBRE REINFORCED COMPOSITE PANELS WITH ALL EDGES SIMPLY-SUPPORTED UNDER BIAXIAL LOADING
Ref. 5	ESDU80023	1995	BUCKLING OF RECTANGULAR SPECIALLY ORTHOTROPIC PLATES
Ref. 6	[MSC]	2012	MSC Nastran 2012. Quick Reference Guide
Ref. 7	[MSC]	-	OVERVIEW OF CLASSICAL LAMINATION THEORY

1.5 DIFFERENCES AMONG VERSIONS

None, this user’s manual corresponds to de first version of the program.

2 PROGRAM EXECUTION

2.1 WINDOWS/PC VERSIONS

The Visual Basic executable program *ANCOite.exe* (as well as an electronic copy of this manual and the program examples) is located in the folder:

```
\\multivac\InfoCalculo\Soft\ANCOlite
```

To execute the program, the user has to click twice on the Visual Basic executable program.

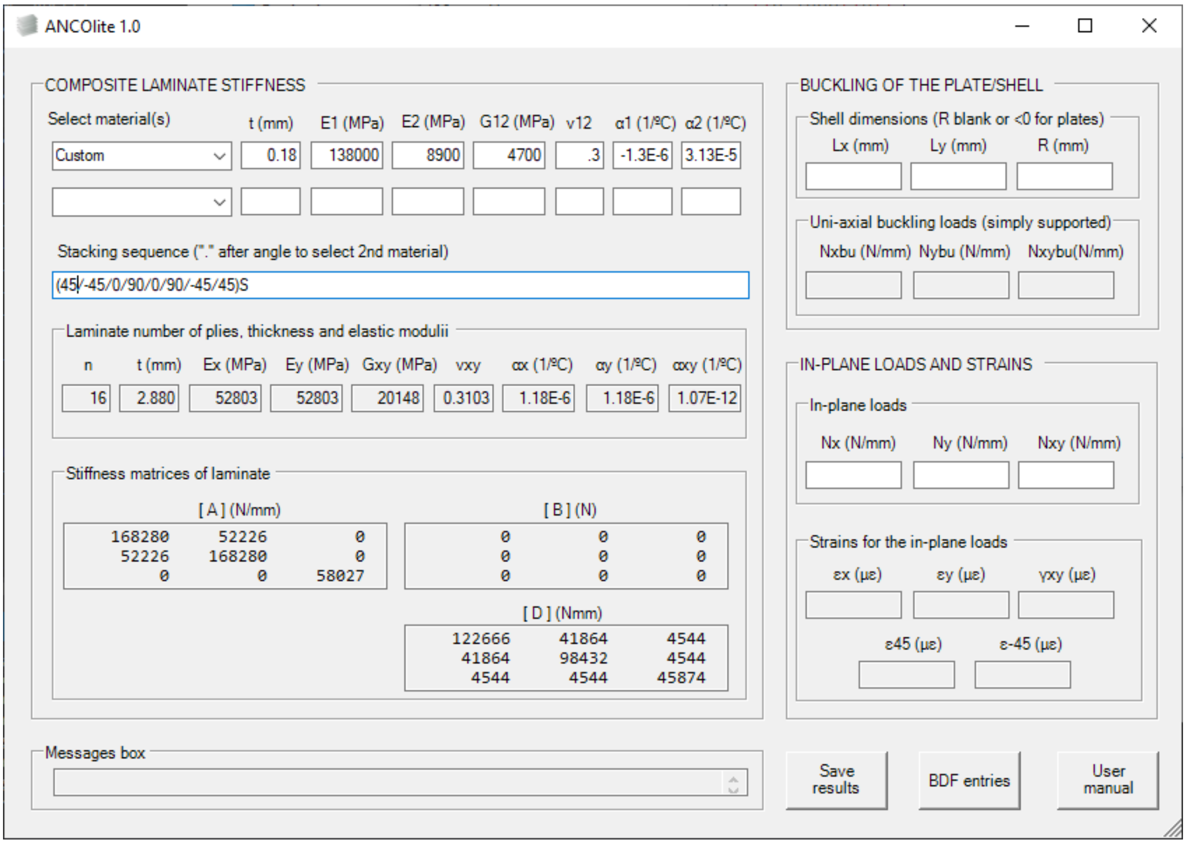


Figure 1. View of GUI for ANCOlite 1.0

3 INPUT DATA

The following input data are required for an execution of *ANCOlite*...

Mandatory input data:

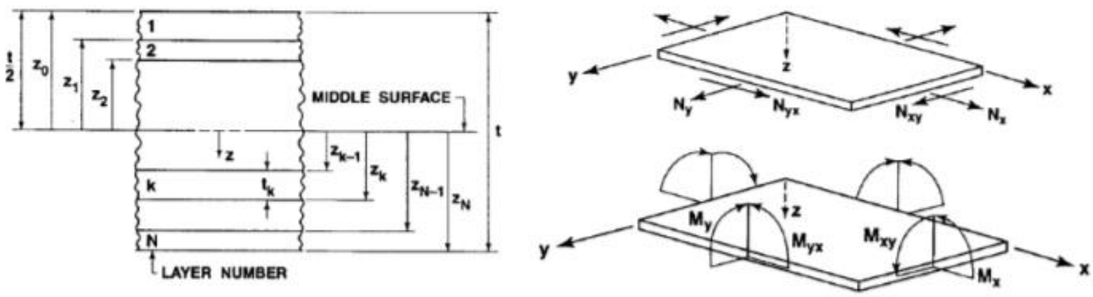
- ✓ Properties of the ply : thickness (*CPT*) and in-plane elastic moduli E_1, E_2, G_{12}, ν_{12}
- ✓ Stacking sequence of ply material and angle defining the laminate

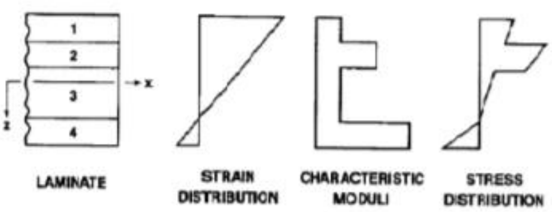
Optional input data:

- ✓ Coefficients of thermal expansion of the material α_1, α_2
- ✓ Plate/shell dimensions for calculation of buckling onsets on uni-axial loads
- ✓ In plane load forces for calculation of strains in symmetric laminates

Physical magnitudes shall be expressed preferably using the subset of units and derived-units of in the International Units system that is the standard for structural analysis: Newtons (N) for force values, milimeters (mm) for length values and Celsius degrees (ºC) for temperatures. Numerical results will be correct if some other units systems are used consistently, but presentation of resulting outputs might look awkward due to the conventions adopted for decimal digits and units labels declared on the software interface.

Sign conventions for input data and output results are those common in literature on the Classical Laminate Theory, e.g. Jones' (Ref. 2).





$$\begin{bmatrix} N_x \\ N_y \\ N_{xy} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} \begin{bmatrix} \varepsilon_x^o \\ \varepsilon_y^o \\ \gamma_{xy}^o \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{bmatrix}$$

$$\begin{bmatrix} M_x \\ M_y \\ M_{xy} \end{bmatrix} = \begin{bmatrix} B_{11} & B_{12} & B_{16} \\ B_{12} & B_{22} & B_{26} \\ B_{16} & B_{26} & B_{66} \end{bmatrix} \begin{bmatrix} \varepsilon_x^o \\ \varepsilon_y^o \\ \gamma_{xy}^o \end{bmatrix} + \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{bmatrix}$$

Figure 2. Basic equation of the Classical Laminate Theory, from Jones

3.1 WINDOWS PC/GUI

The introduction of the input data for several use cases is illustrated with some screenshots...

Choose material in list or fill your own values for E_1, E_2, G_{12} and ν_{12} Thermal expansion coefficients α_1, α_2 are optional inputs

Input the stacking sequence of your laminate, with the stacking angles as integer numbers, i.e. with no decimal digits

Output results

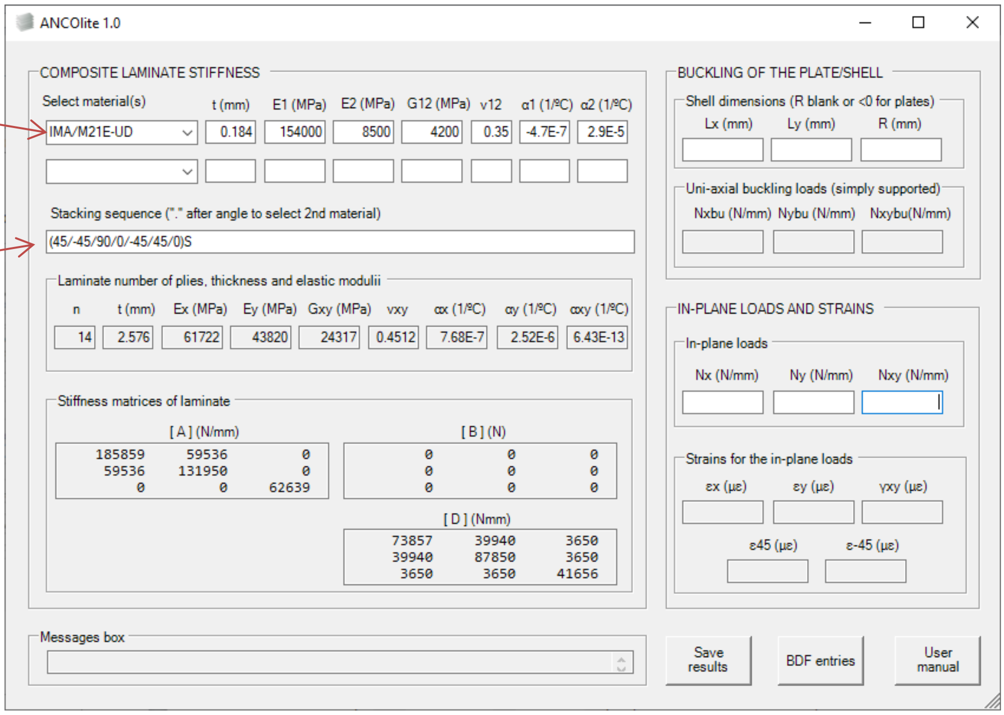


Figure 3. ANCOlite's execution on a mono-material laminate

Choose also a second material in list or fill your own values for E_1, E_2, G_{12} and ν_{12}

Input the stacking sequence (angles) of your laminate, marking the plies made of the second material with a dot (decimals separator) after the angle

If plate/shell dimensions are given, buckling loads are calculated

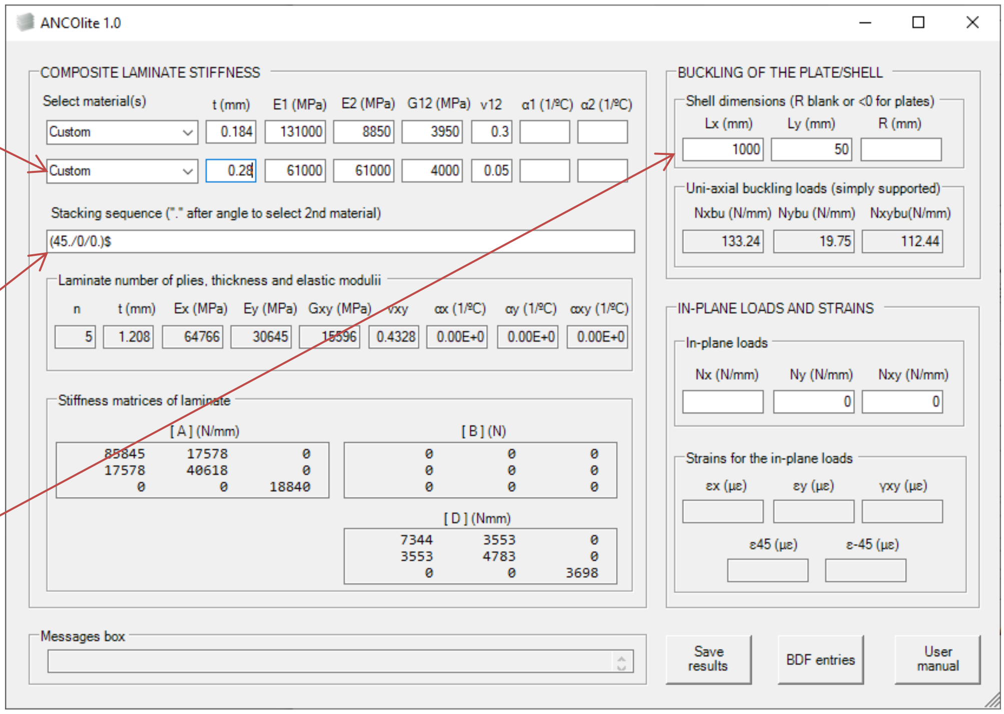


Figure 4. ANCOlite's execution on a bi-material laminate, requesting buckling loads

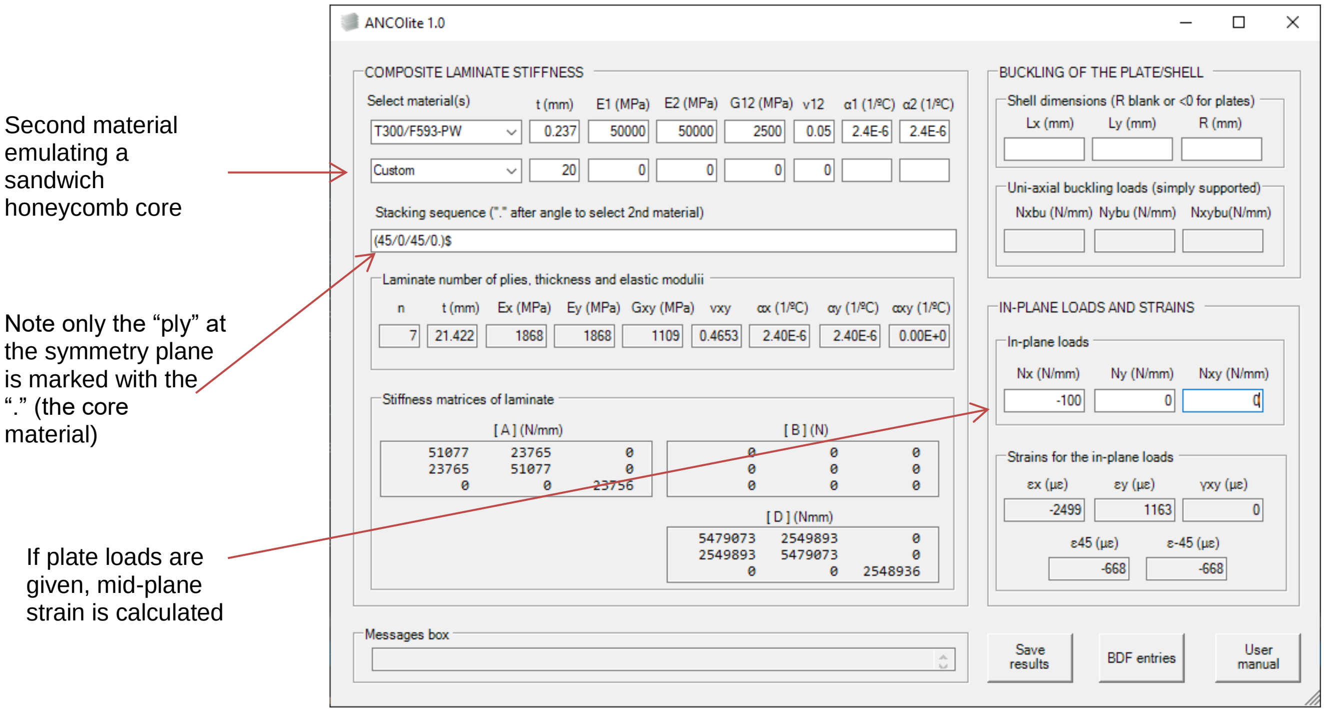


Figure 5. ANCOlite’s execution emulating and sandwich material, requesting faces strains

Input data can be only introduced in the software by means of the interactive graphical form under the Windows OS. Input data cannot be recorded in an input file for reproducing the analysis in a later use just by opening that file. It has not considered necessary as the amount of inputs to fill in the form is short and the output results file write an echo of the input just after the header lines (thus the user has the chance to save a complete record of the execution).

Format of stacking sequence

The following formatting rules apply for defining the stacking sequence of the laminate in *ANCOlite*:

- ✓ Stacking sequence of angles shall be enclosed by brackets, i.e. the “(“ symbol opens the sequence and “)” closes the sequence. Only one level of brackets is valid (nested brackets are not recognized). Square brackets “[“ and “]” are accepted for the same purpose.
- ✓ Lamination angles of the plies shall be expressed in sexagesimal degrees (º) as integer numbers, separated by slash “/” characters
- ✓ Angles defined as an integer numbers indicate that the ply is made of the first material in the selection list, whereas a dot . character after the number mark the ply is made of the 2nd material
- ✓ Accepted conventions for final markers:
    - o An integer number after the closing bracket “)” shall repeat the precedent sublaminate times that number
    - o Use an end “S” character for applying symmetry from the resulting sublaminate and a “$” symbol for symmetry with the plane of symmetry at the last ply of the sublaminate

Valid examples of stacking sequences:

(45/-45/90/0)			
(45/-45/90/0)S	=	(45/-45/90/0/0/90/-45/45)	
(45/-45/90/0)$	=	(45/-45/90/0/90/-45/45)	
(45/-45/90/0)2	=	(45/-45/90/0/45/-45/90/0)	
[45/-45/90/0]2S	=	(45/-45/90/0/45/-45/90/0/0/90/-45/45/0/90/-45/45)	
(45/-45/90/0)2$	=	(45/-45/90/0/45/-45/90/0/90/-45/45/0/90/-45/45)	
(45./0/0.)$	=>	Angles:	(45/0/0/0/45)
		Materials:	(2^{nd}/1^{st}/2^{nd}/1^{st}/2^{nd})

ANCOlite's buttons

The lower-right corner of the GUI includes buttons for the following purposes:

- ✓ *Save results* : open a dialog to save a record of the execution (including an echo of the input data) in a plain text file
- ✓ *BDF entries* : open another form with a text box including the *MAT8, PCOMP* and *MAT2* entries required to use this laminated material in FE analyses with *MSC Nastran*
- ✓ User manual: open this manual

4 OUTPUT RESULTS

4.1 WINDOWS PC/GUI

The following calculations are done and results shown in the GUI:

- ✓ Number of plies and thickness of laminate
- ✓ Apparent/equivalent elastic moduli of laminate (in the plane)
- ✓ Coefficients of thermal expansion of the laminate (in the plane)
- ✓ Stiffness matrices of laminate
- ✓ If dimensions of the rectangular plate/shell are given, uniaxial buckling loads (under X-compression, Y-compression and in-plane shear)
- ✓ If in-plane loads are given, strain level seen in the laminate: X-strain, Y-strain, shear strain and also axial strains at 45º and -45º angles

By pushing the “BDF entries” button, a new Windows form is opened showing the *MAT8, PCOMP* and *MAT2* entries required to use this laminated material in FE analyses with *MSC Nastran.*

4.2 OUTPUT FILE

By pushing the “Save results” button, the input data and abovementioned results can be saved in a plain text files (BDF entries not included).

5 APPENDICES

5.1 LIMITATIONS

As explained in the introductory chapter, *ANCOlite* perform the classical laminate analysis on laminates made of fibrous composite materials: calculation of stiffness matrices and apparent/equivalent elastic moduli, coefficients of thermal expansion, mid-plane strain and buckling loads of the rectangular plate/shell. Actual limitations:

- ✓ Pre-loaded materials list and data shall be updated as design datasheets of fibrous composites are published in the Stress Methods Manual (Ref. 1, Vol. 1)
- ✓ Laminates can be built with a maximum of two ply materials
- ✓ Syntax formatting of stacking sequence not allowing nesting or factoring the repetitions of sublaminates except at the most upper level, just outside the closing bracket and before the end symmetry marker
- ✓ Calculation of the mid-plane strain only for laminates without membrane-bending coupling ($[B]=0$) and cases in-plane/membrane loads
- ✓ The solution for the buckling loads of rectangular simply supported flat plates strictly valid only for orthotropic laminates, but reasonably good for quasi-orthotropic (D_{16}, D_{26} sufficiently small)
- ✓ Method for buckling loads of curved shells still not validated
- ✓ No strength analysis is done nor static margins are calculated

5.2 WARNING AND ERROR MESSAGES

The GUI includes a messages box at the bottom of the form, where the tool shows warnings and error messages of interest, e.g.:

- ✓ Error! Input data in first (main) material not valid
- ✓ Error! Input data in second material not valid
- ✓ Error! Ply properties data not valid
- ✓ Error! Stacking sequence not understood
- ✓ Error (strains)! Input data (loads) not valid
- ✓ Error (strains)! Laminate not valid for in-plane stress analysis
- ✓ Error (buckling)! Input data (plate dimensions) not valid
- ✓ Error (buckling)! Input data (shell radius) must be blank or number
- ✓ Error (buckling)! Input data (plate dimensions) must be positive numbers

- ✓ Warning (buckling)! The method used for shear buckling of a curved shell is a non validated approximation
- ✓ Error (buckling)! Number of semiwaves exceeded limit during calculation of compression buckling loads
- ✓ Error (buckling)! Parameters for shear buckling method exceed allowable limits and result may not be accurate
- ✓ Error (buckling)! Some problem occurred during calculation of uni-axial buckling loads (length to width ratio probably too high)

5.3 EXAMPLES

See illustrative examples shown along the main body of this report.

5.4 THEORETICAL BASIS

5.4.1 Classical Laminate Theory

The Classical Laminate Theory is used for calculating the stiffness matrices of the laminate and its equivalent/apparent elastic moduli, as well strain levels under in-plane loads, as found in technical literature, in particular using notation and conventions in Jones' book (Ref. 2).

The apparent/equivalent elastic moduli of the laminate are obtained from its thickness and the *[A]* stiffness matrix. As explained in SMM (Ref. 1) chapter 3401:

$$E_x = 1 / ([A^{-1}]_{11} \cdot t) = (A_{11} – A_{12}^2 / A_{22}) / t$$

$$E_y = 1 / ([A^{-1}]_{22} \cdot t) = (A_{22} – A_{12}^2 / A_{11}) / t$$

$$\nu_{xy} = - E_x \cdot t \cdot [A^{-1}]_{12} = A_{12} / A_{22}$$

$$G_{xy} = 1 / ([A^{-1}]_{66} \cdot t) = A_{33} / t$$

This is not the same approach used in legacy Airbus DS softwares as *ANCOLAB*, where the assembled *[ABBD]* matrix is used instead. For laminates with non-zero terms in the coupling *[B]* matrix, results will not be the same. For *ANCOlite*, it has been considered more useful to express the apparent/equivalent moduli referenced, not to the mid-plane surface of the laminate, but to that(those) surface(s) representing the elastic axis of the laminate (for which *[B]* terms would be zero). A comprehensive analogy to this new approach could be the centre of gravity in beam sections.

5.4.2 Buckling onsets

Buckling loads for compression in the longitudinal direction X and for compression in the transverse/hoop direction Y can be calculated, for the simply supported plate or cylindrical shell, with the closed form solution in ESDU 94007 (Ref. 4). This formulation converges to that shown in literature for rectangular flat plates when the curvature radius *R* tends to infinite.

Shear buckling of flat plates is obtained from graphical solutions presented in ESDU 80023 (Ref. 5). The solution for cylindrical shells is derived by multiplying that for plates by a curvature correction factor (see SMM chapter 3450). This later method is still under development within Airbus DS and still not considered “official” at the date of release of this document.

It is also worth to note that benchmarks done using this methodology and FEM solutions show some significant differences in wide shells ($Ly > Lx$, *R* non zero) subjected to axial X compression. This mismatch is still under investigation.

5.4.3 Nastran BDF entries

Nastran BDF entries for the laminated material are reported per formatting and conventions documented by *MSC software* in Ref. 6.

The material allowable strains for the *MAT8* entry are arbitrarily high dummy values (1.0 , i.e. 1.0E6 με) in this version of the program (material strengths not considered in *ANCOlite* 1.0).

The stiffness terms for *MAT2* entries are calculated from the stiffness matrices of the laminate as defined by *MSC software* in Ref. 7, adapting the sign of the multiplication factors for keeping consistency between conventions used by MSC and those adopted within *ANCOlite* (and Airbus DS in general):

- ✓ Membrane material in *PSHELL* (*MID1*) $G_{ij} = A_{ij} / t$
- ✓ Bending material in *PSHELL* (*MID2*) $G_{ij} = D_{ij} \cdot 12 / t^3$ (*)
- ✓ Membrane-bending coupling material in *PSHELL* (*MID4*) $G_{ij} = - B_{ij} / t^2$

(*) Valid for the default value (blank or 1.0) of the *12T/T**3* field in the *PSHELL* entry.

The solution for transverse shear stiffness (*MID3* in *PSHELL*) is not reported by *ANCOlite*, in absence of transverse shear data of the material. Calculated apparent G_{xz} and G_{yz} transverse shear moduli of the laminate could be used if transverse shear is considered of interest for the particular analysis.

5.5 VALIDATION CASES

5.5.1 Case *ANCOLAB Ejemplo_1*: stiffness matrices (mono-material)

ANCOLAB ouput

```
                                    PROPIEDADES MEDIAS DEL LAMINADO

        Exx          Eyy          Gxy          Nxy          Nyx        Alfax        Alfay       Alfaxy
   -----------  -----------  -----------  -----------  -----------  -----------  -----------  -----------
     52802.652    52802.652    20148.303    0.3103498    0.3103498    1.178E-06    1.178E-06    0.000E+00


                                        MATRIZ DE RIGIDEZ

              MATRIZ A                             MATRIZ B                             MATRIZ D
              --------                             --------                             --------
  168279.875   52225.625       0.000       0.000       0.000       0.000  122665.992   41864.422    4543.844
   52225.625  168279.891       0.000       0.000       0.000       0.000   41864.422   98432.148    4543.842
       0.000       0.000   58027.125       0.000       0.000       0.000    4543.844    4543.842   45874.426
```

ANCOlite ouputs

- Output on GUI: see Figure 1
- Text output

```
MATERIAL
========

Name                 CPT(mm)     E1(MPa)     E2(MPa)    G12(MPa)        v12  α1(1/ºC)  α2(1/ºC)
---------------     --------  ----------  ----------  ----------   --------  --------  --------
Custom                 0.180      138000        8900        4700      0.300   -1.3E-6    3.1E-5


LAMINATE
========

Stacking sequence
-----------------
(45/-45/0/90/0/90/-45/45)S

Number of plies, thickness and apparent moduli
----------------------------------------------

   n     t(mm)     Ex(MPa)     Ey(MPa)    Gxy(MPa)        vxy  αx(1/ºC)  αy(1/ºC)  αxy(1/ºC)
----  --------  ----------  ----------  ----------   --------  --------  --------  --------
  16     2.880       52803       52803       20148     0.3103    1.2E-6    1.2E-6   1.1E-12

Stiffness matrices
------------------

           [A](N/mm)                          [B](N)                          [D](Nmm)
------------------------------    ------------------------------    ------------------------------
    168280     52226         0             0         0         0        122666     41864      4544
     52226    168280         0             0         0         0         41864     98432      4544
         0         0     58027             0         0         0          4544      4544     45874
```

Conclusion : **results match**

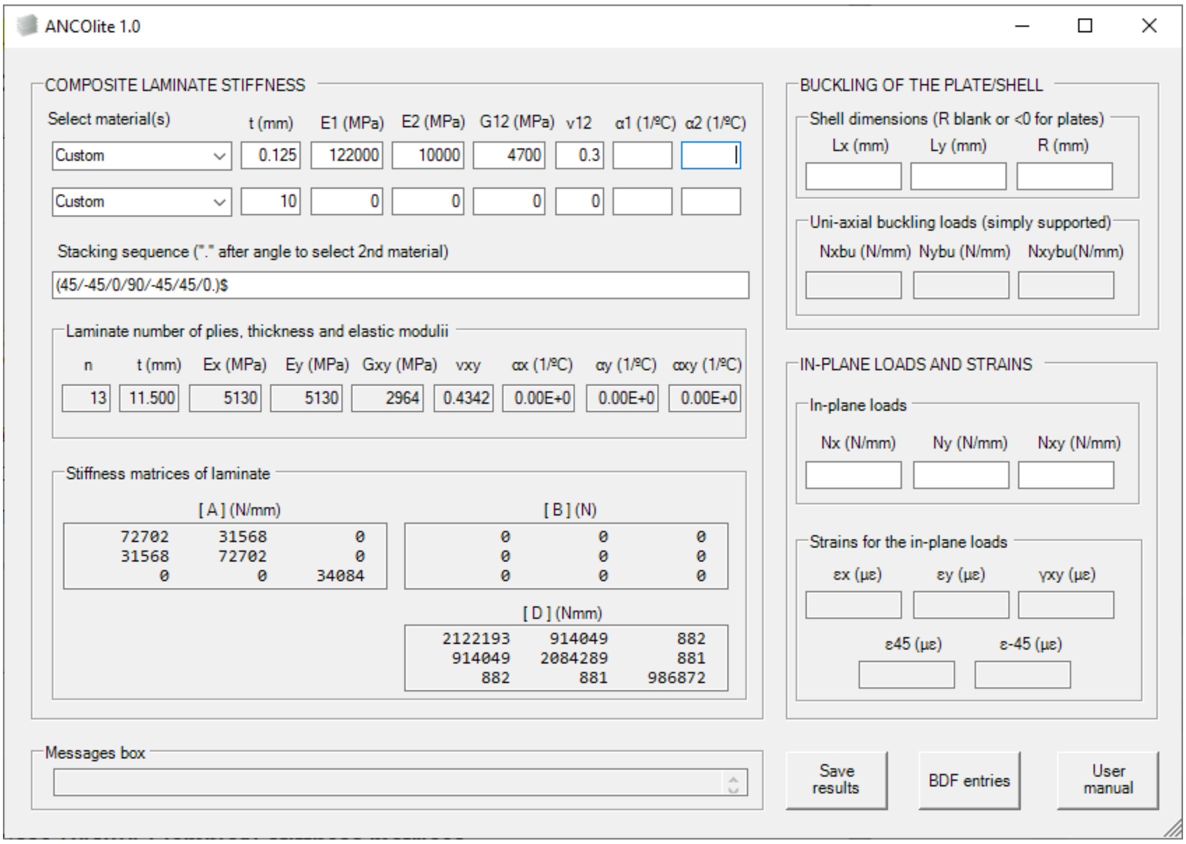
5.5.2 Case *PANSA ejemplo_1*: stiffness matrices (bi-material)

PANSA ouput

```
                                  SANDWICH STIFFNESS MATRIX  (Mean Values)
                                  -------------------------


      INPLANE STIFFNESS MATRIX                    COUPLING MATRIX                     BENDING STIFFNESS MATRIX
 -------------------------------------  -------------------------------------  -------------------------------------
   72701.6      31567.5         0.0          0.0          0.0          0.0       2122192.0     914049.0        881.5
   31567.5      72701.7         0.0          0.0          0.0          0.0        914049.0    2084289.5        881.0
       0.0          0.0     34084.1          0.0          0.0          0.0           881.5        881.0     986871.9

                    Ex           Ey           Exy        Nux      Nuy           Dx           Dy           Dxy
                -----------  -----------  -----------  -----    -----     -----------  -----------  -----------
                     5130.0       5130.0       2963.8   0.43     0.43       1721342.9    1690599.5     986871.9
```

ANCOlite ouputs

- Output on GUI



- Text output

```
MATERIAL
========

Name             CPT(mm)      E1(MPa)     E2(MPa)    G12(MPa)       v12   α1(1/ºC)   α2(1/ºC)
---------------  --------  ----------  ----------  ----------  --------   --------   --------
Custom              0.125      122000       10000        4700     0.300    0.0E+0     0.0E+0
Custom             10.000           0           0           0     0.000    0.0E+0     0.0E+0


LAMINATE
========
```

```
Stacking sequence
-----------------
(45/-45/0/90/-45/45/0.)$

Number of plies, thickness and apparent moduli
----------------------------------------------

   n     t(mm)      Ex(MPa)     Ey(MPa)    Gxy(MPa)       vxy  αx(1/ºC)  αy(1/ºC)  αxy(1/ºC)
----  --------  ----------  ----------  ----------  --------  --------  --------  --------
  13    11.500        5130        5130        2964    0.4342    0.0E+0    0.0E+0    0.0E+0

Stiffness matrices
------------------

           [A](N/mm)                          [B](N)                          [D](Nmm)
------------------------------    ------------------------------    ------------------------------
     72702     31568         0             0         0         0       2122193    914049       882
     31568     72702         0             0         0         0        914049   2084289       881
         0         0     34084             0         0         0           882       881    986872
```

Conclusion : **results match**

5.5.3 Case *ANCOLAB Ejemplo_2*: mid-plane strain

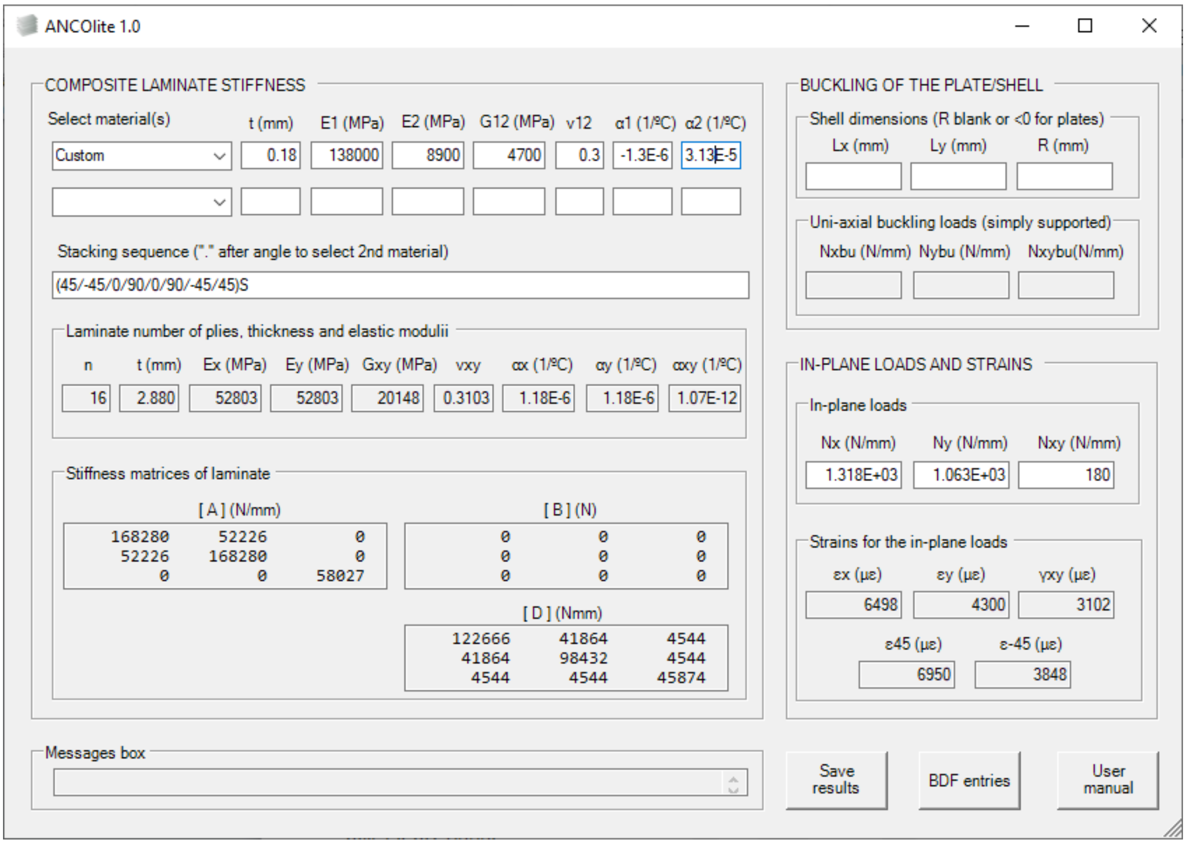
Same laminate as *Ejemplo_1* in ANCOLAB

ANCOLAB ouput

```
                                                    LOAD CASES
                                                  ==============
 Case: 1

   Nx   = 1.318E+03                    Mx   =  0.00                Temp=      0.00

   Ny   = 1.063E+03                    My   =  0.00

   Nxy =   180.                        Mxy =  0.00

   Defx   =  6.500E-03                 Kpx =    0.00

   Defy   =  4.300E-03                 Kpy =    0.00

   Defxy =  3.100E-03                  Kpxy =   0.00
```

ANCOlite ouputs

- Output on GUI



- Text output

```
MEMBRANE STRAINS FOR THE APPLIED LOADS
======================================

  Nx(N/mm)     Ny(N/mm)    Nxy(N/mm)     µeps_x     µeps_y    µgam_xy     µeps45    µeps-45
----------   ----------   ----------   --------   --------   --------   --------   --------
   1318.00      1063.00       180.00       6498       4300       3102       6950       3848
```

<u>Conclusion</u> : **results match** (*)

(*) Differences in strain on the 4th significant digit, probably due to the fact that applied loads in *ANCOLAB* where given as strain (while *ANCOlite* needs load forces/flows, and used the printed output of the former, losing precision with the truncated decimals, as input)

5.5.4 Case *PANPA ej_panpa_5*: plate buckling

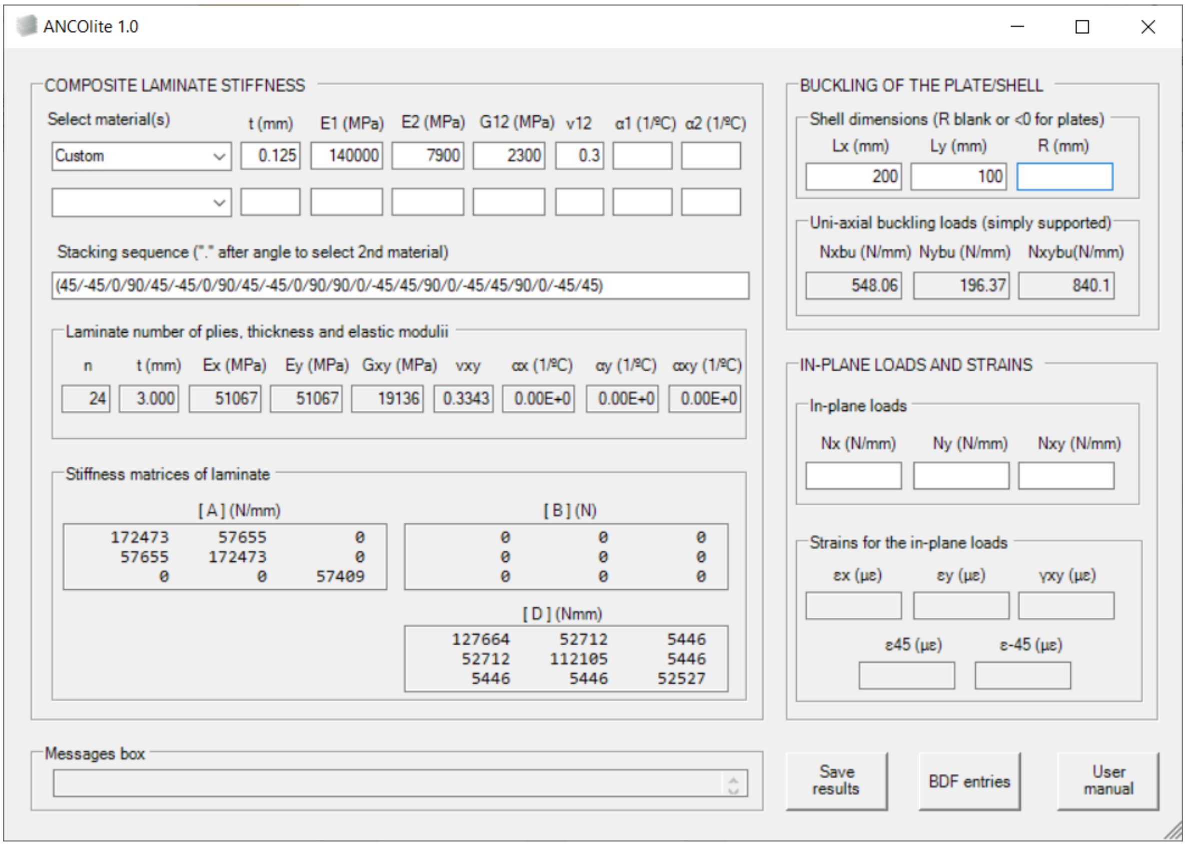
PANPA example *ej_panpa_5* was edited to remove rotational constraint at edges, stiffeners and to apply uni-axial compression and in-plane shear load cases. Record of this execution saved as *ej_panpa_5_ss-uniaxial-loads.dat/out*.

PANPA ouput

```
      A In Plane Stiffness Matrix                   B Coupled Stiffness Matrix                 D Flexural Stiffness Matrix

   172473.3      57655.3          0.0          0.0          0.0          0.0          127664.3      52711.9       5445.8
    57655.3     172473.5          0.0          0.0          0.0          0.0           52711.9     112104.9       5445.8
        0.0          0.0      57409.0          0.0          0.0          0.0            5445.8       5445.8      52527.2


       EX           EY          EXY          NXY          NYX                             DX           DY          DXY
    51066.7      51066.7      19136.3        0.334        0.334                       102879.0      90340.4      52527.2



                      Applied Loads (Flows)                               Buckling Loads (Flows)
         ---------------------------------------------  ---------------------------------------------
  LOAD         FX                  FY                FXY         FX                  FY                FXY
  CASE    Y=0.   Y= 100.00    X=0.   X= 200.00             Y=0.   Y= 100.00    X=0.   X= 200.00               NOX NOY      RF
    1    -100.0   -100.0       0.0      0.0        0.0    -546.6   -546.6       0.0      0.0        0.0      2   1     5.466
    2       0.0      0.0    -100.0   -100.0        0.0       0.0      0.0    -196.0   -196.0        0.0      1   1     1.960
    3       0.0      0.0       0.0      0.0      100.0       0.0      0.0       0.0      0.0      789.8      3   1     7.897
    4       0.0      0.0       0.0      0.0     -100.0       0.0      0.0       0.0      0.0     -880.5      2   1     8.804
```

ANCOlite ouputs

- Output on GUI



- Text output

```
MATERIAL
========

Name              CPT(mm)     E1(MPa)     E2(MPa)    G12(MPa)        v12   α1(1/ºC)  α2(1/ºC)
---------------  --------  ----------  ----------  ----------  --------  --------  --------
Custom              0.125      140000        7900        2300     0.300   0.00E+0   0.00E+0


LAMINATE
========

Stacking sequence
-----------------
(45/-45/0/90/45/-45/0/90/45/-45/0/90/90/0/-45/45/90/0/-45/45/90/0/-45/45)

Number of plies, thickness and apparent moduli
----------------------------------------------

   n      t(mm)     Ex(MPa)     Ey(MPa)    Gxy(MPa)       vxy  αx(1/ºC)  αy(1/ºC)  αxy(1/ºC)
----   --------  ----------  ----------  ----------  --------  --------  --------  --------
  24      3.000       51067       51067       19136    0.3343   0.00E+0   0.00E+0   0.00E+0

Stiffness matrices
------------------

           [A](N/mm)                         [B](N)                         [D](Nmm)
------------------------------    ------------------------------    ------------------------------
    172473     57655         0             0         0         0        127664     52712      5446
     57655    172473         0             0         0         0         52712    112105      5446
         0         0     57409             0         0         0          5446      5446     52527


BUCKLING OF THE SIMPLY SUPPORTED PLATE/SHELL
============================================

    Lx(mm)      Ly(mm)       R(mm)    Nx(N/mm)    Ny(N/mm)   Nxy(N/mm)
----------  ----------  ----------  ----------  ----------  ----------
       200         100         N/A      548.06      196.37      840.10
```

<u>Conclusion</u> : **results match** reasonably well …

- Stiffness matrices and equivalent moduli of laminate match perfectly
- Regarding buckling loads:

Case	***PANPA***	***ANCOlite***
X-compression	-547	-548
Y-compression	-196	-196
Positive in-plane shear	790	840
Negative in-plane shear	880	-840

There is perfect match for cases with axial compression, but a 6% error for shear cases (non-conservative side in case of positive shear). This difference is probably explained by the non-zero terms corresponding to bending-torsion coupling in the *[D]* stiffness matrix of this laminate. The “fast” buckling methods incorporated to *ANCOlite* are strictly valid only for orthotropic laminates. It is expected the error in the prediction is greater as greater are D_{16} and D_{26} terms of the laminate stiffness matrix (relative to terms in *[D]* diagonal).

5.5.5 Case shell buckling

Reserved for future revisions. Validation left for after the fine tuning of the methodology is done.